

Applications of Linear Algebra

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Main message

Linear algebra (matrices, vectors, etc) is used in many contexts.

We will get a taste of how matrices and vectors come up in:

- Image processing
- Graphs
- Quantum computation

Image processing

Puzzle

One of these pictures requires a lot less storage than the other one.

(Source: skimage Python library.)



Can you guess which one?

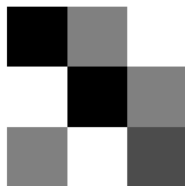
Digital pictures = matrices

A digital black and white picture = matrix with entries in $[0, 1]$.

Example

$$\begin{bmatrix} 0 & 0.5 & 1 \\ 1 & 0 & 0.5 \\ 0.5 & 1 & 0.3 \end{bmatrix}$$

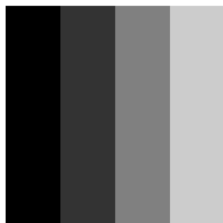
$\rightsquigarrow$



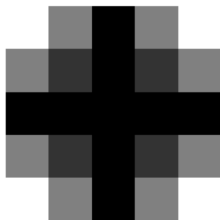
Intuition for image compression

How many numbers do we need to store the following pictures?

$$\begin{bmatrix} 0 & 0.25 & 0.5 & 0.75 \\ 0 & 0.25 & 0.5 & 0.75 \\ 0 & 0.25 & 0.5 & 0.75 \\ 0 & 0.25 & 0.5 & 0.75 \end{bmatrix} \rightsquigarrow$$



$$\begin{bmatrix} 1 & 0.5 & 0 & 0.5 & 1 \\ 0.5 & 0.25 & 0 & 0.25 & 0.5 \\ 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0.25 & 0 & 0.25 & 0.5 \\ 1 & 0.5 & 0 & 0.5 & 1 \end{bmatrix} \rightsquigarrow$$



Low rank matrices

Notation:

- $\mathbb{R}^n = n \times 1$ column vectors
- $\mathbb{R}^{m \times n} = m \times n$ matrices with real entries.

Recall

Rank of $A \in \mathbb{R}^{m \times n}$ = number of linearly independent rows (or columns)

Rank-one matrix

If $A \in \mathbb{R}^{m \times n}$ has rank one, then $A = uv^T$ for some $u \in \mathbb{R}^m, v \in \mathbb{R}^n$.

Low-rank matrix

If $A \in \mathbb{R}^{m \times n}$ has rank k , then

$$A = UV^T$$

for some $U \in \mathbb{R}^{m \times k}, V \in \mathbb{R}^{n \times k}$.

Compression via singular value decomposition

Singular value decomposition (SVD)

Suppose $A \in \mathbb{R}^{m \times n}$. Then A can be factored as

$$A = UDV^T$$

for some $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$, $D \in \mathbb{R}^{m \times n}$ such that

- D is diagonal with non-negative entries,
- $U^T U = I_m$ and $V^T V = I_n$.

Compressed version of A

Suppose the diagonal entries of D are $d_{11} \geq d_{22} \geq \dots$.

Keep only the k largest ones, that is, approximate A with

$$U[:, : k] D[:, k, : k] V[:, : k]^T$$

Notation: $U[:, : k]$ = first k columns of U . Likewise for the others.

Connection between SVD and eigendecomposition

Suppose $A \in \mathbb{R}^{m \times n}$ with SVD $A = UDV^T$.

The eigendecompositions of $A^T A \in \mathbb{R}^{n \times n}$ and $AA^T \in \mathbb{R}^{m \times m}$ are

$$A^T A = V(D^T D)V^T$$

and

$$AA^T = U(DD^T)U^T.$$

Observe

$D^T D \in \mathbb{R}^{n \times n}$ and $DD^T \in \mathbb{R}^{m \times m}$ are diagonal with entries

$$d_{11}^2, d_{22}^2, \dots$$

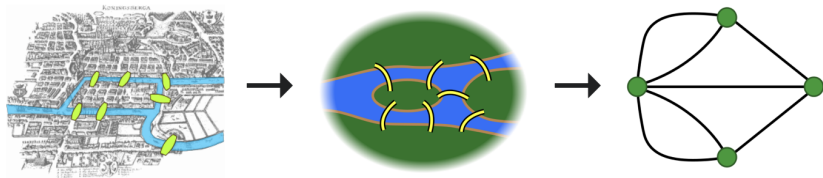
Those are the eigenvalues of $A^T A$ and AA^T .

Graphs and networks

Graphs

Ancient problem: the seven bridges of Königsberg

Devise a walk through the city that would cross each of the seven bridges exactly once.

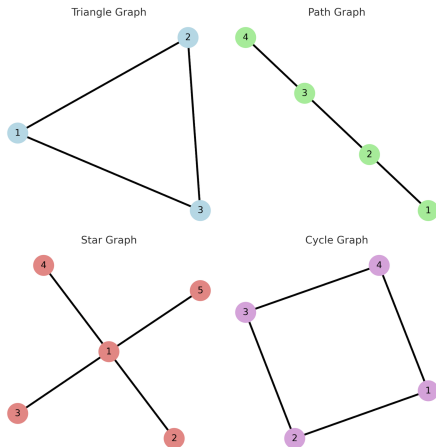


Euler solved this problem in 1735 and created graph theory.

Undirected graph (aka simple graph)

An undirected graph is a pair (V, E) where V is a finite set of vertices and E is a set of **unordered pairs** of elements of V .

Examples



Graphs and matrices

Suppose a graph (V, E) has vertices labeled $1, \dots, n$.

Adjacency matrix

This is a $n \times n$ matrix A with entries 0, 1 entries indicating the vertices joined by edges.

Adjacency matrix for previous graphs (triangle, path, star, cycle):

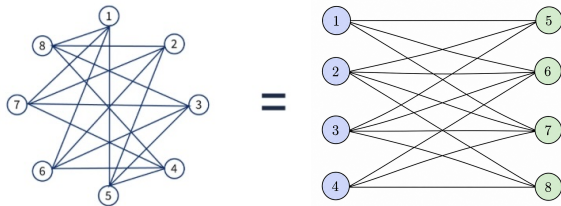
$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Bipartite graphs

A graph $G = (V, E)$ is **bipartite** if we can partition $V = V_1 \cup V_2$ so that every edge connects a vertex in V_1 to a vertex in V_2 .

Example



Can you relate this property to the adjacency matrix?

Graphs and matrices

Suppose a graph (V, E) has vertices labeled $1, \dots, n$.

Laplacian matrix

$$L_{ij} = \begin{cases} \deg(i) & \text{if } i = j \\ -1 & \text{if } i, j \text{ are adjacent} \\ 0 & \text{otherwise.} \end{cases}$$

Laplacian matrix of the star and cycle:

$$\begin{bmatrix} 4 & -1 & -1 & -1 & -1 \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$$

Interesting fact

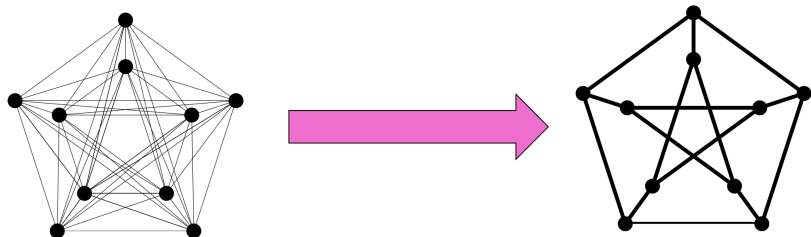
Number of connected components of $G = \dim$ of nullspace of L .

What else can we do with these matrices?

Short answer: a lot.

An interesting application of the Laplacian is *sparsification*: approximate a graph with a sparse one.

Example



To sparsify a graph: use the eigendecomposition of its Laplacian.

Coloring problems

Vertex coloring

Color the vertices of a graph so that no two adjacent vertices have the same color.

Chromatic number of a graph: minimum number of colors needed.

Four color problem

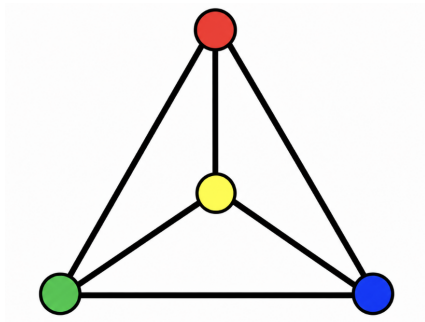
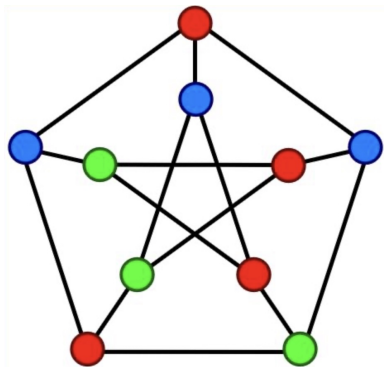
Four colors suffice to color the regions of any map so that no two adjacent regions have the same color.

Equivalent statement: a *planar graph* has chromatic number less than or equal to four.

Alternative definition of vertex coloring

Partition the columns of the adjacency matrix so that any two columns in the same part do not have any ones in common entries.

Some colored graphs

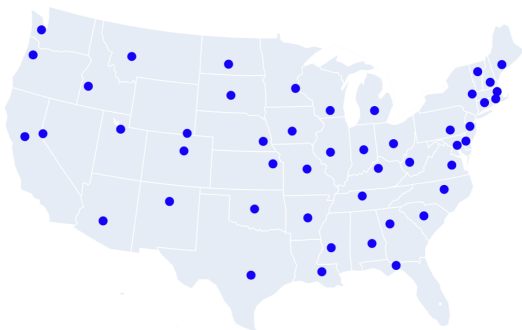


Traveling salesperson problem

Set of locations and distances between every pair of locations.
Find the shortest tour (path that visits every location exactly once).

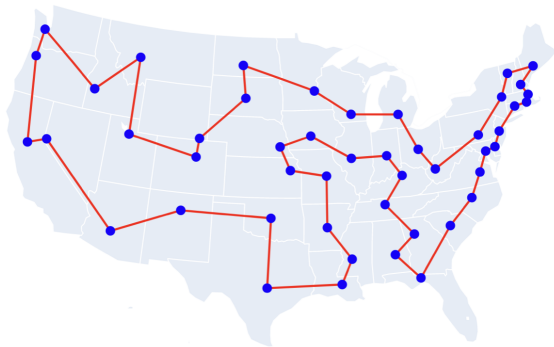
Example

- Consider the 48 state capitals in the continental US.
- What is the fastest route to visit all of them?



Some TSP history

- TSP problems have been documented since the 1850s
- TSP is a famous, difficult, and extensively studied problem.
- Computational breakthrough in 1950s:
 - 1954: Dantzig, Ford, and Fulkerson solved 48-city TSP



Some TSP history

1962: Proctor and Gamble offered \$10,000 prize for a 33-city TSP.



One of the winners was Prof. Gerald Thompson from CMU.

Stable marriage problem

Stable matching problem

Find a *stable* matching between the elements of two equally sized sets given an ordering of preferences for each element.

A matching is *not stable* if there are two unmatched that prefer each other over the element to which they are matched.

A bit of history

- In 1962 Gale and Shapley showed that there is always a stable matching and gave an algorithm to find it.
- In 2012 Shapley and Roth won the Nobel Prize in Economics for using the stable matching problem for market design.

Quantum computing

Picture from 1927 Solvay Conference on Physics



Source: Wikipedia.

Bits and qubits

The **bit** is the fundamental concept of (classic) computation.

Quantum computation is built upon the analog concept of **qubit**.

A bit can be in one of two states: 0 or 1.

A qubit can be in one of the **computational basis** states

$$|0\rangle := \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle := \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

or in a **superposition** of these two states:

$$\alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbb{C}^2$$

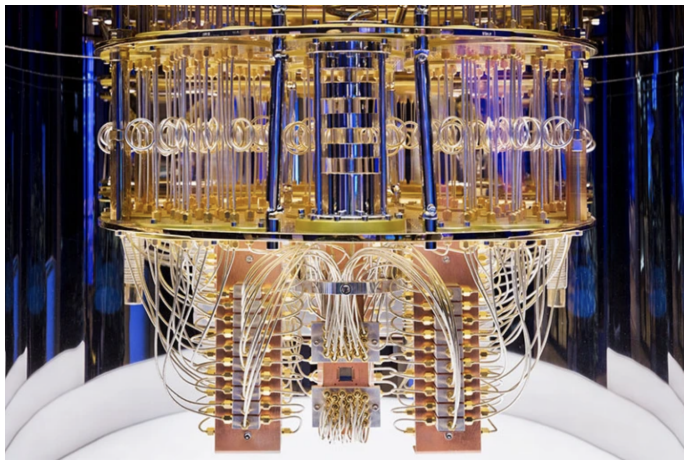
for $\alpha, \beta \in \mathbb{C}$ with $|\alpha|^2 + |\beta|^2 = 1$.

When we measure (observe) the above state

We observe 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

Do quantum computers exist?

Yes. This is a picture of real quantum computer with 100 qubits:



Source

"First quantum computer to pack 100 qubits enters crowded race,"
Nature, Nov 2021.

Largest quantum computer (as of 2026)



Source

Caltech News, October 2025.

Popular quantum algorithms

Quantum search (Grover's algorithm)

Problem: search for a solution to a problem in a set of N elements.

- Classical search algorithms need $\mathcal{O}(N)$ operations.
- Grover's algorithm can do it in $\mathcal{O}(\sqrt{N})$ operations.

Discrete Fourier transform (DFT)

Transform $(x_0, \dots, x_{N-1}) \in \mathbb{C}^N$ to $(y_0, \dots, y_{N-1}) \in \mathbb{C}^N$ via

$$y_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} x_k e^{2\pi i j k / N} \Leftrightarrow y = Fx \text{ for } F_{jk} = \frac{e^{2\pi i j k / N}}{\sqrt{N}}.$$

For $N = 2^n$:

- Best classical algorithm computes DFT in $\mathcal{O}(n2^n)$ operations.
- Quantum Fourier transform can be computed in $\mathcal{O}(n^2)$ quantum operations.

Complex numbers

The complex numbers $\mathbb{C}$ are an extension of the real numbers $\mathbb{R}$.

A complex number has a *real* and an *imaginary* part

$$a + bi$$

where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

To add complex numbers, just add their real and imaginary parts:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

To multiply, expand and use that $i^2 = -1$:

$$\begin{aligned}(a + bi)(c + di) &= ac + adi + bci + bdi^2 \\ &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Fundamental Theorem of Algebra (Gauss 1799)

Every polynomial of positive degree with complex coefficients has at least one complex root.

Complex numbers

Suppose $z = a + bi$.

Conjugate

The *conjugate* of z is

$$z^* = a - bi.$$

Observe that $zz^* = a^2 - b^2i^2 = a^2 + b^2$.

Modulus

The *modulus* (or length) of z is

$$|z| = \sqrt{zz^*} = \sqrt{a^2 + b^2}.$$

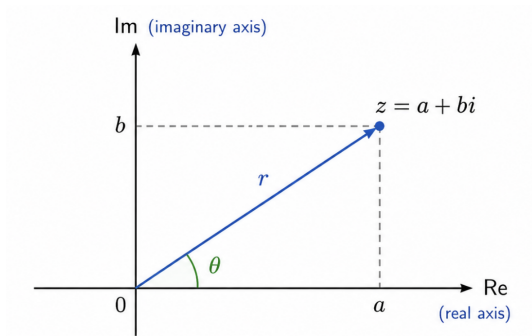
Euler's formula

For $\theta \in \mathbb{R}$

$$e^{i\theta} = \cos(\theta) + i \sin(\theta).$$

Complex numbers

Suppose $z = a + bi$. We can visualize z as follows:



The number z has the following *polar representation*

$$z = |z|e^{i\theta}$$

where $\theta \in [0, 2\pi]$ is such that $\cos(\theta) = a/|z|$ and $\sin(\theta) = b/|z|$.

Visualizing qubits on the Bloch sphere

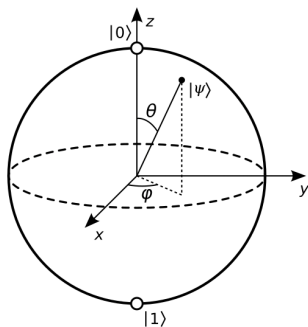
Suppose the state of a qubit is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

Since $|\alpha|^2 + |\beta|^2 = 1$ we can rewrite $|\psi\rangle$ as

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right)$$

for some $\theta \in [0, \pi]$, $\phi, \gamma \in [0, 2\pi]$.

The state $|\psi\rangle$ can be visualized on the **Bloch sphere**:



Qubit operations

The state of a qubit can be seen as an element of $\mathbb{C}^2$:

$$\alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbb{C}^2.$$

The analogue of bit operations, are qubit operations.

The classical NOT gate transforms bits: $0 \rightarrow 1$ and $1 \rightarrow 0$

Quantum NOT gate: $\alpha|0\rangle + \beta|1\rangle \rightarrow \beta|0\rangle + \alpha|1\rangle$.

Quantum NOT gate in matrix form:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightsquigarrow X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

Some quantum gates

Hadamard gate ('square-root of NOT' gate)

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Pauli matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The qubit operations defined by the above gates can be visualized in the Bloch sphere.

Observe

Each of H, X, Y, Z has eigenvalues -1 and 1 .

Intuition for the Bloch sphere

Suppose

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right).$$

Then (after some calculations)

$$|\psi\rangle\langle\psi| = \frac{1}{2} (I + \sin(\theta) \cos(\phi) X + \sin(\theta) \sin(\phi) Y + \cos(\theta) Z)$$

Special cases

- $|\psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightsquigarrow |\psi\rangle\langle\psi| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \frac{1}{2} (I + Z).$
- $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightsquigarrow |\psi\rangle\langle\psi| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} (I + X).$
- $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \rightsquigarrow |\psi\rangle\langle\psi| = \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} = \frac{1}{2} (I + Y).$

Some popular qubit operations

Suppose $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. The transformations

$$|\psi\rangle \mapsto X|\psi\rangle, |\psi\rangle \mapsto Z|\psi\rangle, |\psi\rangle \mapsto H|\psi\rangle,$$

can be shown as follows:

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{X} \longrightarrow \beta|0\rangle + \alpha|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Z} \longrightarrow \alpha|0\rangle - \beta|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{H} \longrightarrow \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Other qubit operations

The Hadamard and Pauli matrices transform the state of a qubit.

In general, the state of a qubit can evolve via a *unitary* transformation. That is, a transformation of the form

$$|\psi\rangle \mapsto U|\psi\rangle$$

where U is a *unitary matrix* defined next.

Hermitian conjugate and unitary matrices

Suppose U is a 2×2 complex matrix.

- The *Hermitian conjugate* $U^\dagger$ is the matrix obtained by transposing and conjugating the entries of U .
- We say that U is unitary if $UU^\dagger = U^\dagger U = I$.

Which of the following matrices are unitary?

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Multiple qubits and entanglement

Two classical bits can be in four possible states:

$$00, 01, 10, 11.$$

A two qubit system has four computational basis states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

The state of two qubits can be in any superposition of them

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

provided $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$.

Bell state (aka EPR pair)

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This state is *entangled*.

Big difference: n bits versus n qubits

A state of n bits is a vector in $\mathbb{R}^n$. It can be any of the 2^n vectors

$$b_1 b_2 \cdots b_n$$

where each $b_i \in \{0, 1\}$.

By contrast, a state of n qubits is a vector in $\mathbb{C}^{2^n} = (\mathbb{C}^2)^{\otimes n}$. It can be a superposition (linear combination) of the 2^n basis vectors

$$|b_1 b_2 \cdots b_n\rangle$$

where each $b_i \in \{0, 1\}$.

However, when we measure the state of n qubits, we get a sequence of n bits.

Some references

- Szeliski, “Computer Vision: Algorithms and Applications”
- Benjamin, Chartrand, and Zhang, “The Fascinating World of Graph Theory”
- Nielsen and Chuang, “Quantum Computation and Quantum Information”